



IoT Platform for Real-Time Monitoring & AI Assistance

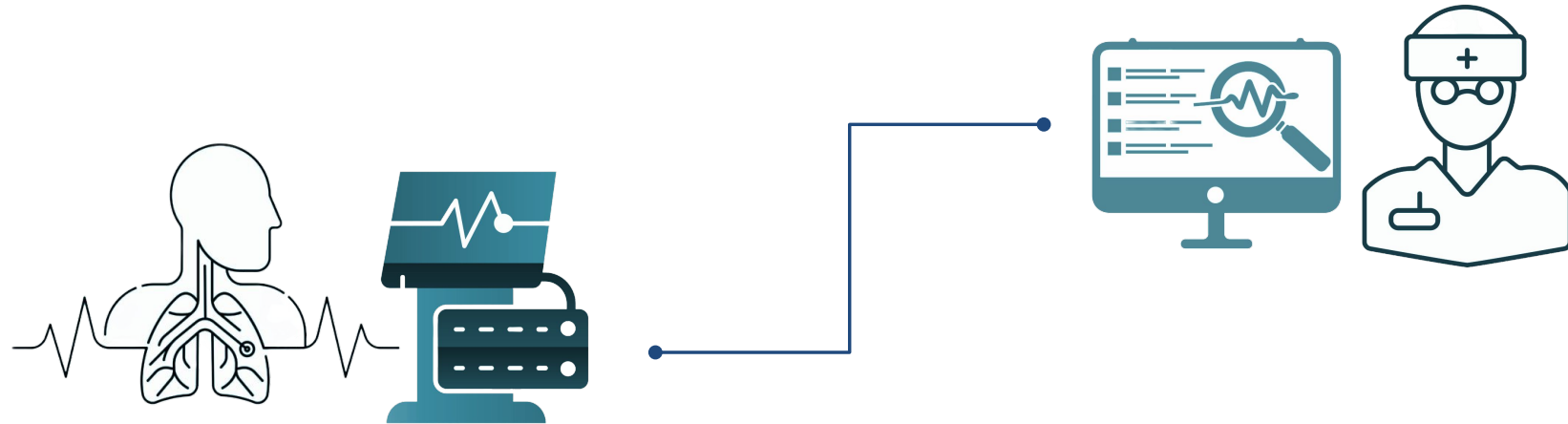
From Sensors to Assistant - An Unified Health Monitoring Solution

Acossi Giorgio - Castagna Marco

A Smarter Way to Monitor and Assist

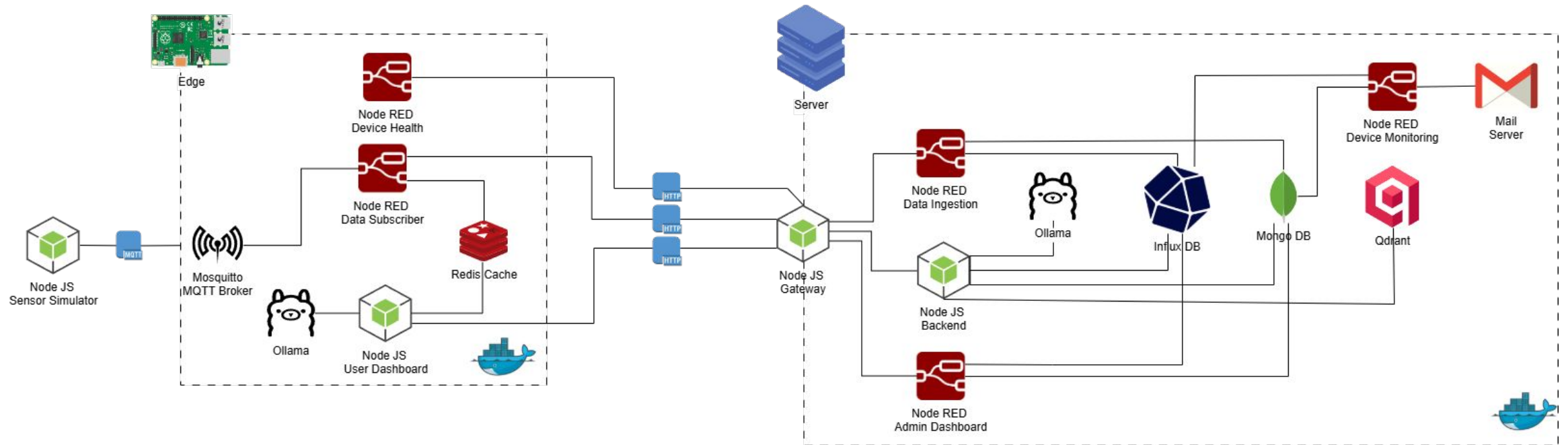
This **platform** acts as a bridge between patients, sensors, and clinicians. It continuously monitors **vital signs** and makes **remote supervision** easier and more reliable for medical experts.

It also provides intuitive **dashboards** and an **AI assistant** capable of interpreting the data, explaining what is happening, and suggesting the most appropriate actions to the patient.

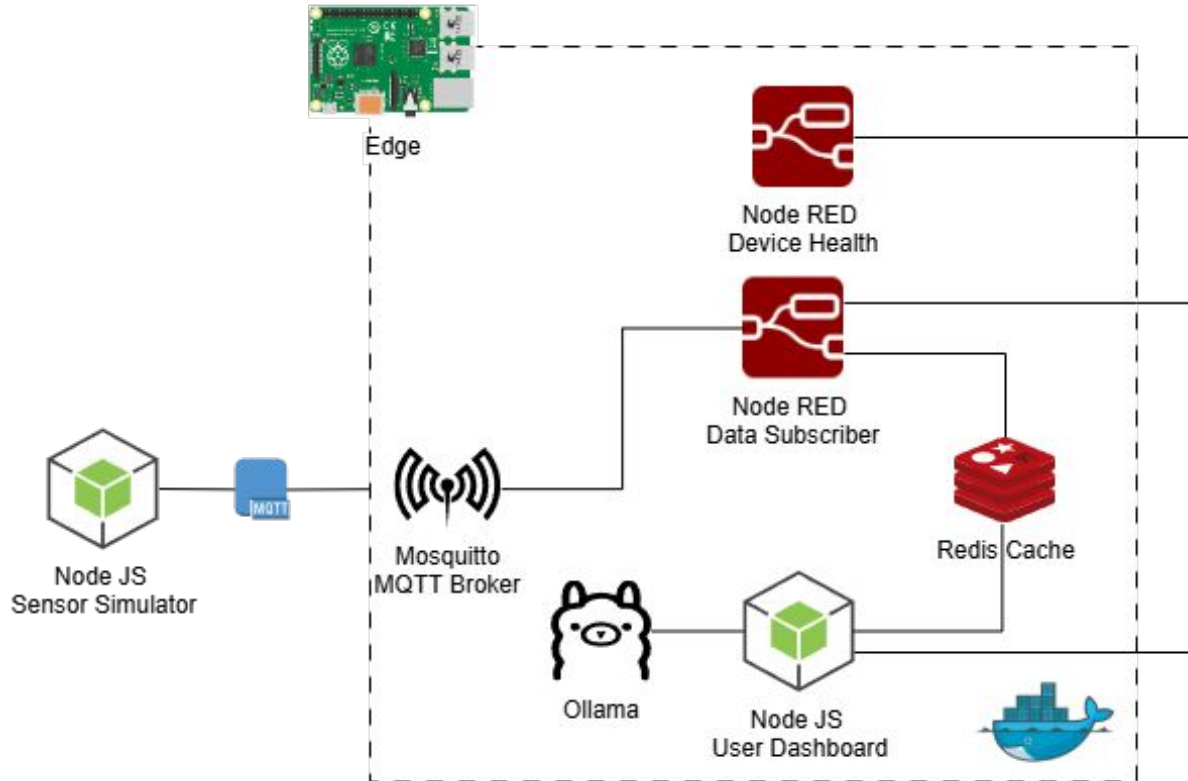


Architecture

Hybrid Approach



Edge Architecture Overview



Data Acquisition

- Sensor simulator
- MQTT broker for edge-level data ingestion

Data Processing & Monitoring

- Device health flow
- Data subscriber flow
- Redis cache for local data storage
- Patient vitals dashboard

AI Assistance

- AI Assistant Chat
- Edge LLM for local filtering and safety

Reliability

- Offline mode

Cloud Architecture Overview

Data Storage

- Time-series storage - InfluxDB
- Application data database - MongoDB
- Vector search index - Qdrant

Data Processing Flows

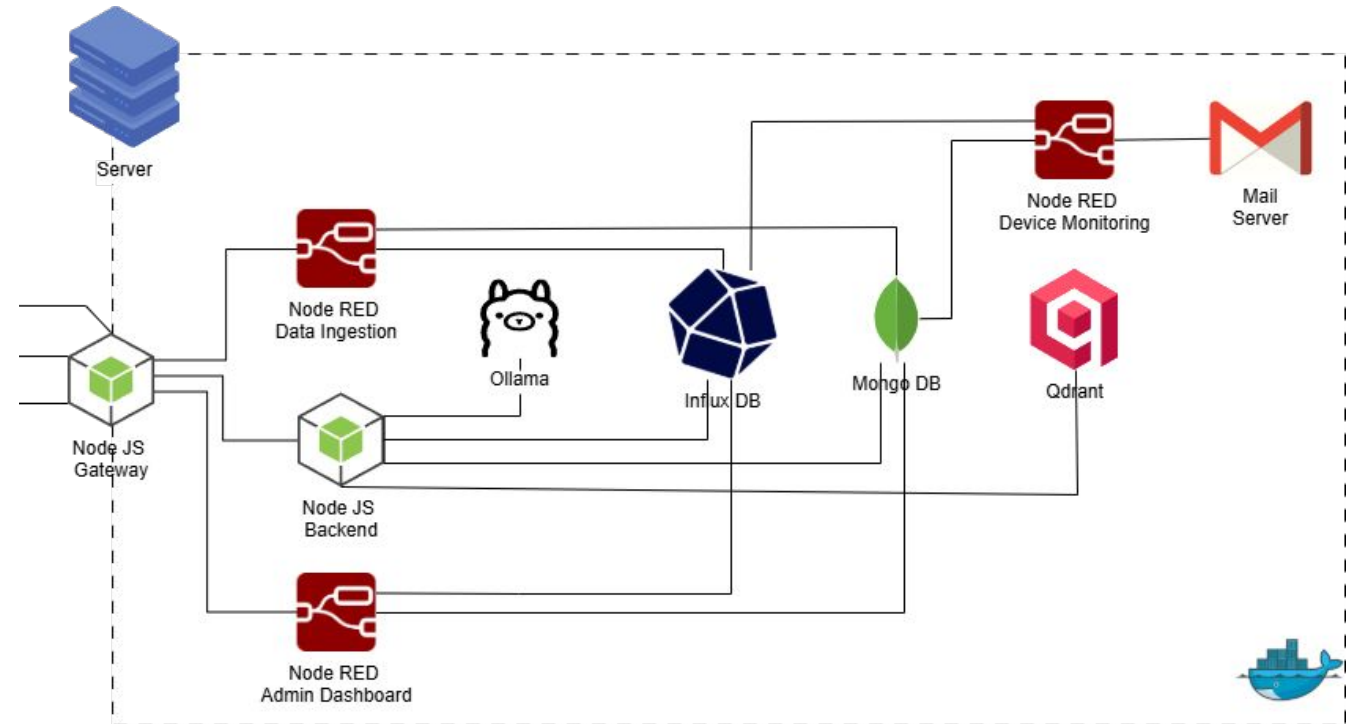
- Data ingestion flow
- Device monitoring flow
- Mail Server for notifications

Dashboards & APIs

- API Gateway and auth
- API Backend
- Admin dashboard

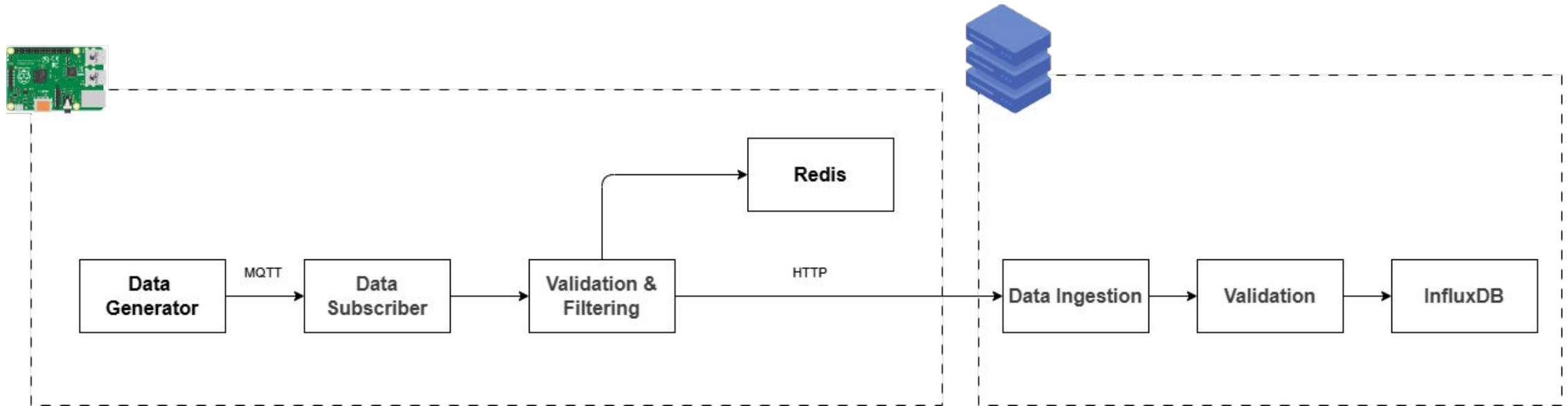
AI Assistant

- Chat (RAG) Pipeline
- Data Analytics Pipeline



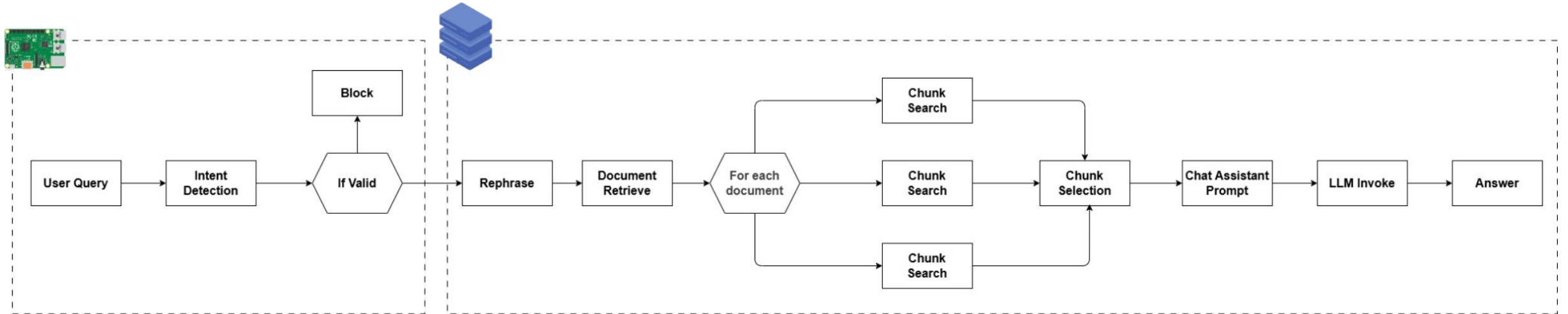
Core Workflow

IoT Data Processing & Ingestion



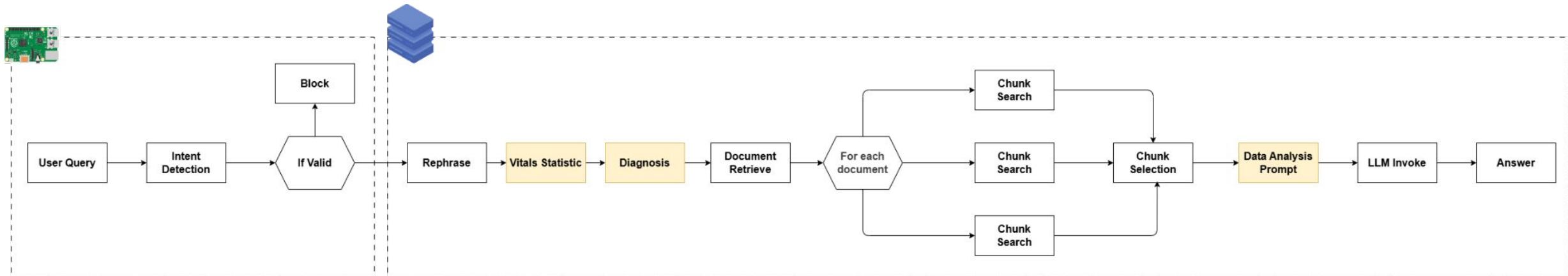
Core Workflow

Chat

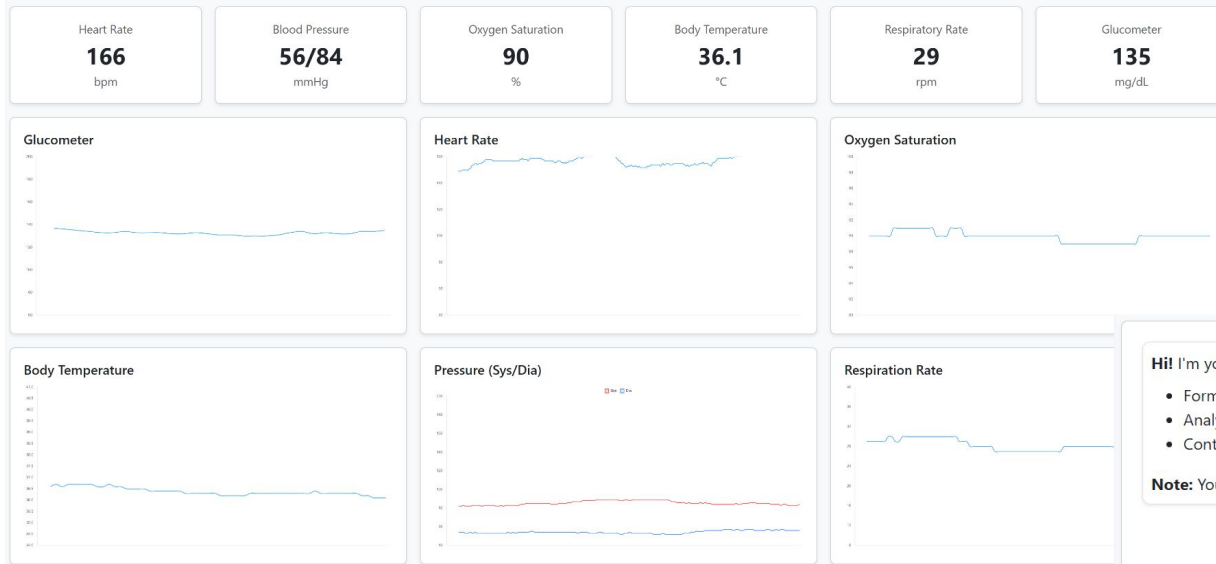


Core Workflow

Data Analysis



Patient Dashboard



Hi! I'm your clinical assistant. I can help you to:

- Formulate and clarify clinical problems
- Analyze your vital data
- Contextualize symptoms and measurements

Note: You can use the **green button** with the chart icon to enable or disable the analysis of your biological data for each query.

I think I have a slight fever. What should I do ?

Could you analyze my vital signs and tell me if there is any problem ? 



Expert Dashboard

Devices			
Search			
Device Id	Name	Status	Last Update
device-1	Device-1		2025-12-02T22:29:30.981Z
device-2	Device-2		-
device-3	Device-3		-
device-4	Device-4		-

Sessions				
Search				
Id	Device Id	Created At	Updated At	Status
692f541f7b76d3f4cd8db7b87	device-1	2025-12-02T21:03:27.201Z	2025-12-02T21:03:27.201Z	
692f53e17b76d3f4cd8db7b85	device-1	2025-12-02T21:02:25.433Z	2025-12-02T21:03:25.870Z	
692f126c3ab92063c48e2507	device-1	2025-12-02T16:23:08.460Z	2025-12-02T16:27:37.269Z	
692f10233ab92063c48e2506	device-1	2025-12-02T16:13:23.677Z	2025-12-02T16:23:06.825Z	
692f0b423ab92063c48e2505	device-1	2025-12-02T15:52:34.336Z	2025-12-02T16:13:23.674Z	
692f0b273ab92063c48e2504	device-1	2025-12-02T15:52:07.228Z	2025-12-02T15:52:33.436Z	
692ec61f505b94edfe6651c9	device-1	2025-12-02T10:57:35.199Z	2025-12-02T15:52:07.223Z	
692ec5aa505b94edfe6651c7	device-1	2025-12-02T10:55:38.849Z	2025-12-02T10:57:34.354Z	
692ec4cd505b94edfe6651c6	device-1	2025-12-02T10:51:57.716Z	2025-12-02T10:55:38.008Z	
692ec3bf505b94edfe6651c4	device-1	2025-12-02T10:48:27.131Z	2025-12-02T10:51:56.577Z	
692e1535a262cd39bc591bea	device-1	2025-12-01T22:22:45.094Z	2025-12-02T10:48:27.128Z	
692e11ff386e1f7db71b6970	device-1	2025-12-01T22:09:03.277Z	2025-12-01T22:22:43.804Z	

Session Details																											
Person ID: 692f67047d6c77643bb779d3																											
Glucose vitals, glucose (mg/dL) 200 180 160 140 120 100 80 60 23:25 23:26 23:27 23:28 23:29 time SPO2 vitals, spo2 (%) 100 95 90 85 80 23:25 23:26 23:27 23:28 23:29 time Heart Rate vitals, heart_rate (bpm) 160 140 120 100 80 60 40 23:25 23:26 23:27 23:28 23:29 time Blood Pressure vitals, blood pressure_syst (mmHg) vitals, blood pressure_dia (mmHg) 200 180 160 140 120 23:25 23:26 23:27 23:28 23:29 time Body Temperature vitals, body_temperature (°C) 41 40 39 38 37 36 35 34 23:25 23:26 23:27 23:28 23:29 time Respiratory Rate vitals, respiratory_rate (breaths/min) 40 35 30 25 23:25 23:26 23:27 23:28 23:29 time <tr><th colspan="6">Interactions</th></tr> <tr><th>Question</th><th>Rephrased Question</th><th>Answer</th><th>Chunks</th><th>Vitals</th><th>Diagnosis</th><th>Type</th><th>Created At</th></tr> <tr><td>Could you analyze my vitals data and tell me if there is any problem?</td><td>Could you analyze my vital data and tell me if there is anything wrong?</td><td> Your vital signs indicate serious abnormalities requiring immediate medical attention: Respiratory Rate: 29 breaths/minute (normal: 12-20) - significantly elevated, suggesting respiratory distress. Oxygen Saturation: 80% (normal: 95-100%) - critically low, indicating hypoxemia. Heart Rate: 153 bpm (normal: 60-100) - severely elevated tachycardia. Systolic Blood Pressure: 56 mmHg (normal: 90-120) - dangerously low, indicating hypotension. The combination of low oxygen saturation, very high heart rate, and critically low blood pressure is life-threatening. This pattern is consistent with severe respiratory failure, possible heart attack (myocardial infarction), or dangerous heart rhythm abnormalities. These signs require emergency evaluation and intervention. Do not delay seeking urgent medical care. </td><td> - Oxford Handbook of General Practice.pdf (Pg.1083) - Mayo Clinic Family Health Book.pdf (Pg.2162) - Mayo Clinic Family Health Book.pdf (Pg.2166) - Mayo Clinic Family Health Book.pdf (Pg.2160) - Mayo Clinic Family Health Book.pdf (Pg.2133) - Mayo Clinic Family Health Book.pdf (Pg.2128) - Mayo Clinic Family Health Book.pdf (Pg.2142) - Mayo Clinic Family Health Book.pdf (Pg.2126) - Mayo Clinic Family Health Book.pdf (Pg.2127) - Mayo Clinic Family Health Book.pdf (Pg.2139) </td><td> Respiration Rate: Min: 25, Max: 36, Mean: 28.1, Latest: 29 Diastolic Blood Pressure: Min: 81, Max: 85, Mean: 84.6, Latest: 85 SPO2: Min: 81, Max: 91, Mean: 89.9, Latest: 89 Systolic Blood Pressure: Min: 52, Max: 56, Mean: 51.4, Latest: 56 Heart Rate: Min: 149, Max: 159, Mean: 155.8, Latest: 153 Glucose: Min: 139, Max: 151, Mean: 136.7, Latest: 133 Body Temperature: Min: 36.3, Max: 36.7, Mean: 36.4, Latest: 36.3 </td><td> - Respiratory failure - Possible myocardial infarction - Tachyarrhythmia </td><td>dataAnalysis</td><td>2025-12-02T22:27:59.7867</td></tr>						Interactions						Question	Rephrased Question	Answer	Chunks	Vitals	Diagnosis	Type	Created At	Could you analyze my vitals data and tell me if there is any problem?	Could you analyze my vital data and tell me if there is anything wrong?	Your vital signs indicate serious abnormalities requiring immediate medical attention: Respiratory Rate: 29 breaths/minute (normal: 12-20) - significantly elevated, suggesting respiratory distress. Oxygen Saturation: 80% (normal: 95-100%) - critically low, indicating hypoxemia. Heart Rate: 153 bpm (normal: 60-100) - severely elevated tachycardia. Systolic Blood Pressure: 56 mmHg (normal: 90-120) - dangerously low, indicating hypotension. The combination of low oxygen saturation, very high heart rate, and critically low blood pressure is life-threatening. This pattern is consistent with severe respiratory failure, possible heart attack (myocardial infarction), or dangerous heart rhythm abnormalities. These signs require emergency evaluation and intervention. Do not delay seeking urgent medical care.	- Oxford Handbook of General Practice.pdf (Pg.1083) - Mayo Clinic Family Health Book.pdf (Pg.2162) - Mayo Clinic Family Health Book.pdf (Pg.2166) - Mayo Clinic Family Health Book.pdf (Pg.2160) - Mayo Clinic Family Health Book.pdf (Pg.2133) - Mayo Clinic Family Health Book.pdf (Pg.2128) - Mayo Clinic Family Health Book.pdf (Pg.2142) - Mayo Clinic Family Health Book.pdf (Pg.2126) - Mayo Clinic Family Health Book.pdf (Pg.2127) - Mayo Clinic Family Health Book.pdf (Pg.2139)	Respiration Rate: Min: 25, Max: 36, Mean: 28.1, Latest: 29 Diastolic Blood Pressure: Min: 81, Max: 85, Mean: 84.6, Latest: 85 SPO2: Min: 81, Max: 91, Mean: 89.9, Latest: 89 Systolic Blood Pressure: Min: 52, Max: 56, Mean: 51.4, Latest: 56 Heart Rate: Min: 149, Max: 159, Mean: 155.8, Latest: 153 Glucose: Min: 139, Max: 151, Mean: 136.7, Latest: 133 Body Temperature: Min: 36.3, Max: 36.7, Mean: 36.4, Latest: 36.3	- Respiratory failure - Possible myocardial infarction - Tachyarrhythmia	dataAnalysis	2025-12-02T22:27:59.7867
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Simulation Scenario

Chat

 Dashboard

 Chat

Hi! I'm your clinical assistant. I can help you to:

- Formulate and clarify clinical problems
- Analyze your vital data
- Contextualize symptoms and measurements

Type me a question, for example:

- "I don't feel very well, which parameters are out of range?"

I think that i could have some fever, what i could do ?

Take your temperature to confirm a fever (typically defined as 100.4°F/38°C or higher). Rest and drink plenty of fluids to stay hydrated. Monitor your symptoms closely. Seek medical attention immediately if:

- You are under 3 months old with any fever
- Your temperature exceeds 103°F (39.4°C)
- You experience severe symptoms like confusion, stiff neck, shortness of breath, chest pain, or a rash
- Fever lasts more than 3 days without improvement
- You have underlying health conditions like heart disease, diabetes, or weakened immunity

For most adults and older children with mild fever, home care (rest, fluids, over-the-counter pain relievers like acetaminophen or ibuprofen) is sufficient unless symptoms worsen or persist.

✕ Close



|



Simulation Scenario

Data Analysis - Healthy

Dashboard

Chat

Heart Rate

75

bpm

Blood Pressure

76/127

mmHg

Oxygen Saturation

99

%

Body Temperature

36.7

°C

Respiratory Rate

16

rpm

Glucometer

99

mg/dL

Glucometer



Heart Rate



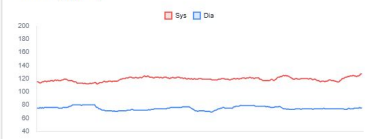
Oxygen Saturation



Body Temperature



Pressure (Sys/Dia)



Respiration Rate



Dashboard

Chat

Hi! I'm your clinical assistant. I can help you to:

- Formulate and clarify clinical problems
- Analyze your vital data
- Contextualize symptoms and measurements

Note: You can use the **green button** with the chart icon to enable or disable the analysis of your biological data for each query.

Could you analyze my vitals data and tell me if there is any problem ?

Your vital signs are all within normal ranges and show consistent stability. Here's a clear breakdown:

- **Respiration Rate:** 17 breaths per minute (normal range: 12-20), with consistent breathing patterns.
- **Blood Glucose:** 93 mg/dL (normal fasting range: 70-99), showing stable levels.
- **Oxygen Saturation:** 97% (normal range: 95-100%), indicating excellent oxygen levels.
- **Blood Pressure:** 122/72 mmHg (normal: systolic <120, diastolic <80), with stable readings.
- **Heart Rate:** 74 bpm (normal resting range: 60-100), consistent with good cardiovascular function.
- **Body Temperature:** 36.6°C (normal range: 36.1-37.2°C), showing no signs of fever or hypothermia.

All measurements demonstrate low variability, confirming a stable physiological state. There are no indications of medical concerns based on these vitals.

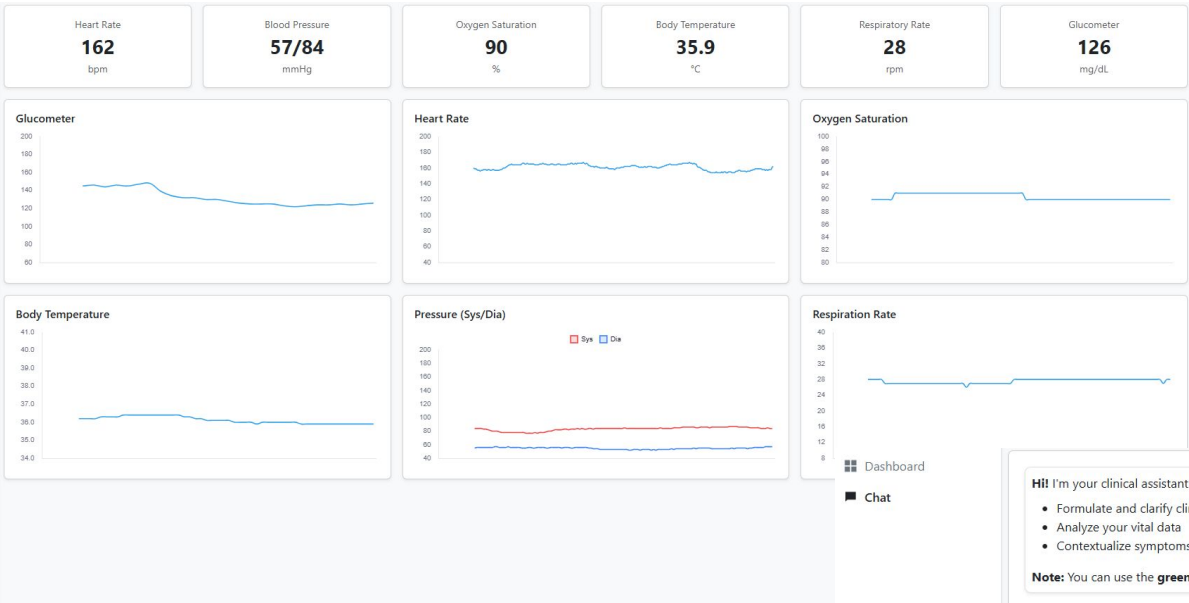
Close



Simulation Scenario

Data Analysis - Hearth Attack

Dashboard
Chat



Close

Dashboard
Chat

Hi! I'm your clinical assistant. I can help you to:

- Formulate and clarify clinical problems
- Analyze your vital data
- Contextualize symptoms and measurements

Note: You can use the **green button** with the chart icon to enable or disable the analysis of your biological data for each query.

Could you analyze my vitals data and tell me if there is any problem ?

Your vital signs show several significant abnormalities that require immediate medical evaluation:

- **Heart rate (160 bpm)** is severely elevated (normal resting range: 60-100 bpm), indicating possible tachycardia.
- **Systolic blood pressure (84 mmHg)** is critically low (normal: 90-120 mmHg), suggesting hypotension.
- **Oxygen saturation (91%)** is below normal (normal: 95-100%), indicating potential oxygen deficiency.
- **Respiratory rate (27 breaths/min)** is elevated (normal resting: 12-20 breaths/min), reflecting increased breathing effort.

These combined findings—high heart rate, low blood pressure, low oxygen, and rapid breathing—are clinical indicators of a serious cardiovascular or respiratory emergency. They align with conditions such as heart attack (myocardial infarction) or life-threatening arrhythmias, as suggested by your algorithm diagnosis. Do not delay seeking emergency medical care.

Close



Future Developments

LLM Fine
Tuning

Graph RAG

Autonomous
Agent

Security
Improvement

ML Driven
Diagnosis

Kubernetes

Real
Device & Data

Questions & Answers

Why did you choose Node.js to build a RAG?

Is it easier to build a dashboard in Node-RED or by your hand?

Could the RAG also have been implemented at the edge?